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| 1(a)(i)   | <p>any <b>two</b> from:<br/> <i>idea that, <u>sister</u> / <u>identical</u>, chromatids need to be present (for mitosis) ;</i><br/> <b>A</b> ref. to 'chromatins' if chromatids noted elsewhere in answer</p> <p>need to, check for errors / avoid mutations (before progressing to mitosis) ;<br/> <b>A</b> cell does not begin mitosis unless S phase checkpoint is passed AW<br/> <b>bod</b> ref. to proofreading</p> <p>(so) daughter cells / cells produced by mitosis, are genetically identical ;<br/> <b>A</b> need to provide each daughter cell with same, DNA / chromosomes<br/> <b>A</b> (so) sister / identical, chromatids can be in separate (daughter) cells<br/> <b>I</b> so genetically identical to (original) parent cell</p> <p><i>idea that, DNA / chromatin / chromosome, too condensed during, mitosis / prophase (for replication to occur) ; ora</i><br/> <b>A</b> DNA cannot unwind (for replication to occur) AW<br/> AVP ; e.g.<br/> DNA inactive during mitosis / no transcription and translation (for histone production)<br/> replication takes longer than prophase so cannot occur in mitosis</p> | 2 |
| 1(a)(ii)  | <div style="display: flex; align-items: center;"> <div style="margin-right: 10px;"> <u>DNA</u> polymerase ;<br/><br/> <u>DNA</u> ligase ; </div> <div style="font-size: 3em; margin-right: 10px;">}</div> <div> <i>either order</i><br/> <b>R</b> DNA lyase </div> </div> <p><b>A</b> correct non-syllabus enzymes e.g. helicase / topoisomerase / DNA primase<br/> <i>spellings must be correct</i></p> <p><i>if no marks gained, allow 1 mark for polymerase <u>and</u> ligase but <b>R</b> <u>RNA</u> polymerase and ligase</i></p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               | 2 |
| 1(a)(iii) | <p>S (phase) / G<sub>1</sub> (phase) ; <b>A</b> S and G<sub>1</sub> (phases) <b>R</b> if G<sub>2</sub> also stated</p> <p>(proteins) need to be complexed with DNA during, replication / S phase ; AW<br/> <i>ecf on G<sub>2</sub> phase</i><br/> <i>e.g. need to be complexed with DNA after, replication / S phase</i></p> <p>AVP ; e.g. (need) histone proteins for DNA /chromosome, coiling / condensation (in mitosis)</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      | 2 |

| Question | Answer                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         | Marks   |                              |   |   |                                                                                                                                           |         |   |                                                                                        |      |   |                                                                 |      |   |                                                                                                                             |      |  |                                                                                                                                                         |  |   |                                                                                                                                              |   |   |                                                                                                                                         |      |   |                                                                                           |   |   |                                              |   |   |
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| 1(b)     | <p>I ref. to large vacuole (not present in dividing plant cells)<br/>any <b>three</b> from: look for relevant <b>ora</b></p> <table border="1"> <tr> <td>1</td><td>no, cell plate / cell wall ;</td><td>A</td></tr> <tr> <td>2</td><td>cleavage furrow present      <i>accept ref. to contractile ring</i><br/>or<br/>cytoplasm / cell (surface) membrane, pinching in / constricting ;</td><td>A, B, D</td></tr> <tr> <td>3</td><td><u>cytokinesis</u> occurring / (cell is in late) <u>telophase</u> ; <b>R</b> cytokines</td><td>B, D</td></tr> <tr> <td>4</td><td>cytoplasm (still), being shared out / dividing / connected ; AW</td><td>B, D</td></tr> <tr> <td>5</td><td>cell (surface) membrane(s) not fully formed (around each cell) ;<br/><b>bod</b> cell membrane(s) is forming (to divide cell)</td><td>B, D</td></tr> <tr> <td></td><td><i>if mp 4 and 5 not gained allow mark for idea of cells still connected / still a single cell / membranes still connected / not divided completely</i></td><td></td></tr> <tr> <td>6</td><td><i>idea of</i>, cell structures / organelles / cytoplasm, not shared out equally ;<br/><b>A</b> (forming daughter) cells are different sizes</td><td>C</td></tr> <tr> <td>7</td><td>nuclear envelope(s) not yet formed (around chromosomes)<br/>or<br/>(two separate) nuclei not visible / nucleus in each part not visible ;</td><td>B, D</td></tr> <tr> <td>8</td><td>(daughter) chromosomes, (still) visible / (still) condensed / not diffuse ;<br/><b>ora</b></td><td>D</td></tr> <tr> <td>9</td><td>spindle fibres, (still), visible / present ;</td><td>D</td></tr> </table> <p><b>Key</b></p> <p>Observation 1      not plant cells = <b>A</b></p> <p>Observation 2      — not two cells = <b>B</b><br/>                                     not identical = <b>C</b><br/>                                     not early interphase = <b>D</b></p> | 1       | no, cell plate / cell wall ; | A | 2 | cleavage furrow present <i>accept ref. to contractile ring</i><br>or<br>cytoplasm / cell (surface) membrane, pinching in / constricting ; | A, B, D | 3 | <u>cytokinesis</u> occurring / (cell is in late) <u>telophase</u> ; <b>R</b> cytokines | B, D | 4 | cytoplasm (still), being shared out / dividing / connected ; AW | B, D | 5 | cell (surface) membrane(s) not fully formed (around each cell) ;<br><b>bod</b> cell membrane(s) is forming (to divide cell) | B, D |  | <i>if mp 4 and 5 not gained allow mark for idea of cells still connected / still a single cell / membranes still connected / not divided completely</i> |  | 6 | <i>idea of</i> , cell structures / organelles / cytoplasm, not shared out equally ;<br><b>A</b> (forming daughter) cells are different sizes | C | 7 | nuclear envelope(s) not yet formed (around chromosomes)<br>or<br>(two separate) nuclei not visible / nucleus in each part not visible ; | B, D | 8 | (daughter) chromosomes, (still) visible / (still) condensed / not diffuse ;<br><b>ora</b> | D | 9 | spindle fibres, (still), visible / present ; | D | 3 |
| 1        | no, cell plate / cell wall ;                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   | A       |                              |   |   |                                                                                                                                           |         |   |                                                                                        |      |   |                                                                 |      |   |                                                                                                                             |      |  |                                                                                                                                                         |  |   |                                                                                                                                              |   |   |                                                                                                                                         |      |   |                                                                                           |   |   |                                              |   |   |
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| 3        | <u>cytokinesis</u> occurring / (cell is in late) <u>telophase</u> ; <b>R</b> cytokines                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         | B, D    |                              |   |   |                                                                                                                                           |         |   |                                                                                        |      |   |                                                                 |      |   |                                                                                                                             |      |  |                                                                                                                                                         |  |   |                                                                                                                                              |   |   |                                                                                                                                         |      |   |                                                                                           |   |   |                                              |   |   |
| 4        | cytoplasm (still), being shared out / dividing / connected ; AW                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                | B, D    |                              |   |   |                                                                                                                                           |         |   |                                                                                        |      |   |                                                                 |      |   |                                                                                                                             |      |  |                                                                                                                                                         |  |   |                                                                                                                                              |   |   |                                                                                                                                         |      |   |                                                                                           |   |   |                                              |   |   |
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